

Quiz (Habanero)

- Q1.** What is the graph of the inequality $y > 2x - 3$?
(A) A line with a slope of 2 and y-intercept -3
(B) A line with a slope of -2 and y-intercept 3
(C) A line with a slope of 2 and y-intercept 3
(D) A solid line with a slope of 2 and y-intercept -3
(E) A dashed line with a slope of 2 and y-intercept -3
- Q2.** Which inequality represents the region below the line $y = -x + 4$?
(A) $y < -x + 4$
(B) $y > -x + 4$
(C) $y \leq -x + 4$
(D) $y \geq -x + 4$
(E) $y = -x + 4$
- Q3.** The temperature in a city is given by $t = -2h + 25$, where t is the temperature in degrees Celsius and h is the number of hours since sunrise. Which inequality represents the temperatures above 20 degrees Celsius?
(A) $-2h + 25 > 20$
(B) $-2h + 25 < 20$
(C) $-2h + 25 \geq 20$
(D) $-2h + 25 \leq 20$
(E) $-2h + 25 = 20$
- Q4.** A river's water level is given by $w = 3d - 2$, where w is the water level in meters and d is the distance from the river's source. Which inequality represents the water levels below 5 meters?
(A) $3d - 2 < 5$
(B) $3d - 2 > 5$
(C) $3d - 2 \leq 5$
(D) $3d - 2 \geq 5$
(E) $3d - 2 = 5$
- Q5.** The graph of the inequality $y \leq \frac{1}{2}x + 2$ has a
(A) solid boundary line
(B) dashed boundary line
(C) vertical boundary line
(D) horizontal boundary line
(E) no boundary line
- Q6.** The region represented by the inequality $y > -\frac{3}{4}x - 1$ is
(A) above the line $y = -\frac{3}{4}x - 1$
(B) below the line $y = -\frac{3}{4}x - 1$
(C) on the line $y = -\frac{3}{4}x - 1$
(D) to the left of the line $y = -\frac{3}{4}x - 1$
(E) to the right of the line $y = -\frac{3}{4}x - 1$
- Q7.** Which of the following is the graph of the inequality $x + 2y > 4$?
(A) a line with a slope of -1/2 and y-intercept 2
(B) a line with a slope of 1/2 and y-intercept -2
(C) a line with a slope of -1/2 and y-intercept -2
(D) a solid line with a slope of -1/2 and y-intercept 2
(E) a dashed line with a slope of -1/2 and y-intercept 2
- Q8.** The inequality $y < 2x - 5$ represents the region
(A) below the line $y = 2x - 5$
(B) above the line $y = 2x - 5$
(C) on the line $y = 2x - 5$
(D) to the left of the line $y = 2x - 5$
(E) to the right of the line $y = 2x - 5$

Open Ended Questions (FRQ)

- Q9.** Graph the inequality $y > \frac{1}{3}x - 2$ and shade the region above the line. Provide the equation of the boundary line.
- Q10.** A company's profit is given by $p = -2c + 150$, where p is the profit in dollars and c is the cost in dollars. Write an inequality to represent the profits above 50. *Provide a graph of the inequality and shade the region above the line.*

Chef's Tips

- Tip 1: Make sure to use a ruler to draw the lines on the graph.
- Tip 2: Use a pencil to shade the region, as it may need to be erased.
- Tip 3: Check that the inequality is in the correct form before graphing.